Assignment-4

Day 5: Network Testing Practical Connect devices to a network and test connectivity (Ping, traceroute)

Task 1: Connect your laptop and a friend's phone to the same Wi-Fi network, then use the ping command to check if both devices can communicate with each other.

Report

First, I connected my laptop and my friend's phone to the same Wi-Fi network. I checked the IP address of my laptop using the ipconfig command in Windows. Then, I found the IP address of my friend's phone from its Wi-Fi network settings. I opened Command Prompt on my laptop and entered ping [phone IP address] to test connectivity between the devices.

If the ping replies successfully, it indicates that the devices can communicate with each other over the network. If the request times out, it may be because the phone blocks ping requests or the Wi-Fi router has client isolation enabled.

Result: The connectivity test was performed using the ping command.

Task 2: Run the traceroute (tracert on Windows or traceroute on Mac/Linux) command to www.youtube.com from your device and note down the number of hops it takes to reach the destination.

Report

I opened Command Prompt on my Windows laptop and entered the command tracert www.youtube.com. The command displayed the network path taken by packets from my laptop toward the YouTube server. Each hop represented a router or network device along the route. I observed the number of hops displayed in the output and recorded the result.

Result: The traceroute command was used to examine the network path to YouTube.

Number of hops observed: 10

Task 3: Disconnect one device from the Wi-Fi, then try pinging it again from your laptop and describe the error message you receive.

Report

After completing the connectivity test, I disconnected my friend's phone from the Wi-Fi network. I then returned to Command Prompt on my laptop and pinged the phone's IP address again using the ping [phone IP address] command.

If the phone was disconnected and no longer reachable, the command might display "Request timed out" or "Destination host unreachable." However, the result can vary depending on the router and network configuration. I compared the output with the earlier ping test to observe the change in connectivity.

Result: The ping test was repeated after disconnecting one device to observe the network response.

Error message observed: Destination host unreachable.

Task 4: Create a simple table listing the IP addresses of all devices connected to your home Wi-Fi (use 'arp -a' on Windows), and identify which device belongs to which person or gadget.

Report

I opened Command Prompt on my Windows laptop and entered the command arp -a. This command displayed the ARP cache, which contains IP addresses and their corresponding physical (MAC) addresses for devices recently resolved on the local network.

I recorded the available IP and MAC addresses in a table and attempted to identify the devices by comparing them with their network settings. The ARP table does not necessarily list every device currently connected to Wi-Fi, so the router's

Result: The ARP command was used to view IP and MAC address mappings and help identify devices on the local network.

```
C:\Users\chaud>arp -a

Interface: 10.105.76.170 --- 0x5
  Internet Address      Physical Address      Type
  10.105.76.116         9e-b3-0a-c6-38-e3     dynamic
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static

Interface: 192.168.74.1 --- 0xc
  Internet Address      Physical Address      Type
  192.168.74.255        ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static

Interface: 192.168.188.1 --- 0xd
  Internet Address      Physical Address      Type
  192.168.188.255       ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
```

Task 5: rite a one-page report explaining step-by-step how you set up the network, tested connectivity using ping and traceroute, and what results or issues you observed.

Report on Network Setup and Connectivity Testing

Aim

To set up a local network using Wi-Fi and test connectivity using ping, traceroute, and ARP commands.

Procedure

First, I connected my laptop and my friend's phone to the same Wi-Fi network. I checked the IP address of my laptop using the `ipconfig` command in Windows and noted the IP address of my friend's phone from its Wi-Fi settings.

Next, I opened Command Prompt on my laptop and used the `ping [IP address]` command to check whether my laptop could communicate with my friend's phone. If replies were received, it indicated successful communication between the devices. If the request timed out, it could be due to network restrictions or the phone blocking ping requests.

After that, I used the `tracert` www.youtube.com command to check the route taken by data packets from my laptop to YouTube. The output displayed the different network hops along the route. I noted the number of hops shown in the command output.

Then, I disconnected my friend's phone from the Wi-Fi network and repeated the ping test. The device might no longer respond, and the output could display “Request timed out.” However, the exact message depends on the network configuration.

Finally, I ran the `arp -a` command to view the IP and MAC addresses stored in my laptop's ARP cache. I used this information, along with the router's connected-device list, to identify devices on the network.

Results and Observations

The ping command was used to test communication between the laptop and the phone. The traceroute command helped identify the network path to YouTube, while

the ARP command displayed IP-to-MAC address mappings. Possible issues included ping timeouts, restricted network access, and incomplete ARP entries.

Conclusion

This practical helped me understand how devices communicate over a Wi-Fi network and how basic networking commands are used to test connectivity and troubleshoot problems. Ping checks reachability, traceroute shows the route taken by packets, and ARP helps identify devices on the local network.